

Computational Methods for Metaoptics: From Full-Wave Simulation to Machine-Learning Inverse Design

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Abstract: Metasurfaces—ultrathin planar structures composed of subwavelength resonators—have emerged as a transformative platform for controlling light at the nanoscale, promising to replace bulky conventional optical components with flat, lightweight, and CMOS-compatible alternatives. The design of metaoptical devices is inherently simulation-driven, as the parameter space is vast and fabrication is costly. This review provides a systematic and comparative analysis of the computational methods that underpin metaoptics: finite-difference time-domain (FDTD) simulation, Fourier optics and angular spectrum propagation, machine-learning and inverse design approaches, and hybrid multi-scale methods that combine these techniques. For each method, we present the mathematical foundations, practical implementation considerations, representative applications across key metaoptics domains—including metalens design, beam shaping, holography, achromatic devices, tunable metasurfaces, sensing, and computational co-design—and critically assess accuracy–cost trade-offs and fundamental limitations. We identify the scalability wall that prevents full-wave simulation of macroscopic devices, the local-phase-approximation breakdown at high numerical apertures, the training-data dependency of neural network surrogates, and the fabrication–simulation gap as the principal open challenges. Finally, we chart emerging directions including fully differentiable end-to-end design pipelines, physics-informed neural networks, and standardized benchmarks for metaoptics simulation, providing a practical roadmap for researchers entering this rapidly evolving field.

1. Introduction

The quest to control light at will has driven centuries of innovation in optics, from Galileo’s telescopes to modern lithographic lenses that pattern integrated circuits with nanometer precision. Conventional refractive and reflective optical elements, however, rely on gradual phase accumulation through bulk propagation, imposing fundamental constraints on size, weight, and system complexity. The emergence of *metamaterials*—engineered composites whose electromagnetic properties arise from subwavelength structuring rather than chemical composition—opened a conceptually new route to light manipulation, most dramatically illustrated by Pendry’s proposal for a perfect lens exploiting negative refraction [1]. Yet three-dimensional metamaterials remained difficult to fabricate at optical frequencies, limiting practical impact.

The transformative leap came with the recognition that a single, ultrathin layer of subwavelength resonators—a *metasurface*—can impart abrupt, spatially varying phase shifts to an incident wavefront, bypassing the need for volumetric propagation entirely. The seminal demonstration by Yu *et al.* [2] of anomalous refraction governed by a generalized Snell’s law,

$$n_t \sin \theta_t - n_i \sin \theta_i = \frac{\lambda_0}{2\pi} \frac{d\Phi}{dx}, \quad (1)$$

where $d\Phi/dx$ is the local phase gradient imposed by the metasurface, established the foundational design principle for flat optics. Subsequent advances replaced lossy plasmonic resonators with high-index dielectric nanostructures [3, 4], enabling diffraction-limited focusing at visible

wavelengths [5] and ushering in the era of high-efficiency *metaoptics*—functional flat optical devices including metalenses, beam shapers, holograms, and spectrometers.

The appeal of metaoptics is multifold. Flat form factors reduce size, weight, and power (SWaP) requirements by orders of magnitude compared to conventional lens stacks. Compatibility with standard semiconductor fabrication processes—including electron-beam lithography (EBL), deep-ultraviolet (DUV) lithography, and nanoimprint lithography—promises scalable, wafer-level mass production [6]. Perhaps most compellingly, a single metasurface can simultaneously control phase, amplitude, and polarization of transmitted or reflected light, enabling multifunctional devices that would require cascaded bulk elements in conventional optics [7, 8].

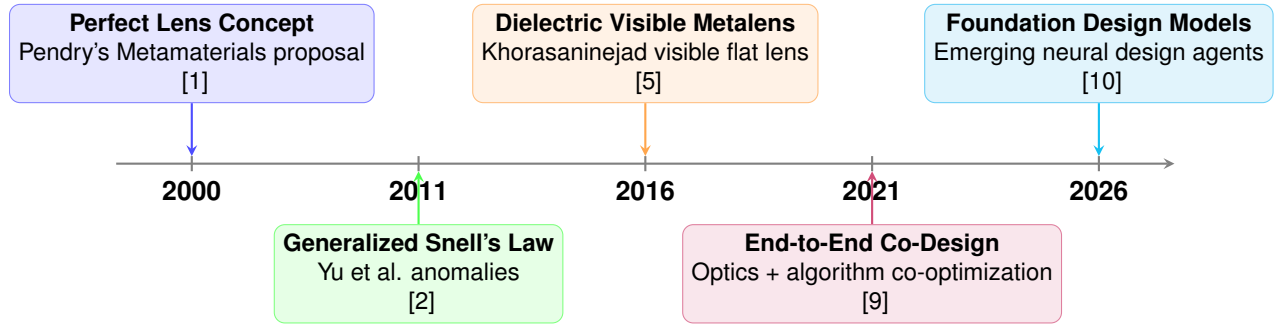


Fig. 1. Timeline of key milestones in the evolution of metaoptics, highlighting the transition from bulky 3D metamaterials to planar metasurfaces and computational optics.

Despite these remarkable capabilities, the design of metaoptical devices remains a formidable computational challenge. A typical metalens comprises 10^6 – 10^9 individual meta-atoms, each described by multiple geometric parameters (shape, diameter, height, period, orientation), operating across a range of wavelengths and incidence angles. The resulting design space is astronomically large, rendering exhaustive experimental exploration impractical. Fabrication turnaround times of days to weeks and costs of hundreds to thousands of dollars per device further incentivize a *simulation-first* design paradigm. Consequently, the choice of computational method fundamentally constrains the complexity, performance, and scale of the metaoptical devices that can be designed.

Several excellent reviews have surveyed metasurface physics and applications [1, 3, 4, 6, 11], and recent perspectives have highlighted the role of machine learning in nanophotonics [10, 12–14]. However, no review has provided a *systematic, cross-cutting comparison* of computational methods—their mathematical foundations, practical trade-offs, and domain-specific suitability—across the full landscape of metaoptics applications. This gap is significant: researchers routinely confront the question of which simulation approach is appropriate for their specific device, and the answer depends critically on device scale, required accuracy, computational budget, and target application.

This review addresses that gap. We provide a comprehensive and comparative analysis of four classes of computational methods for metaoptics: (i) full-wave finite-difference time-domain (FDTD) simulation, (ii) Fourier optics and angular spectrum propagation, (iii) machine-learning and inverse design approaches, and (iv) hybrid multi-scale methods. For each, we present mathematical foundations, implementation considerations, representative applications, and a critical assessment of accuracy–cost trade-offs. We then examine how these methods are deployed across seven major application domains—metalens design, beam shaping, holography, broadband/achromatic devices, tunable metasurfaces, sensing, and computational co-design—before identifying the principal open challenges and emerging directions.

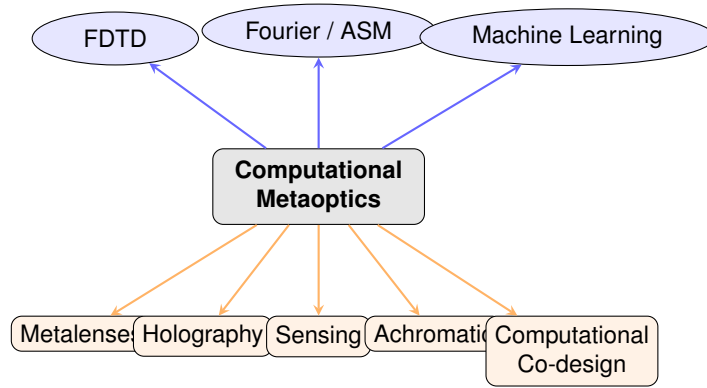


Fig. 2. Schematic of the review scope, mapping core simulation methodologies (blue ellipses) to key application domains (orange rectangles).

The remainder of this article is organized as follows. Section 2 reviews the fundamentals of metasurface design, including phase control mechanisms, material platforms, and the local phase approximation. Section 3 presents the core comparative analysis of simulation methodologies. Section 4 surveys applications through the lens of computational methods employed. Section 5 critically examines limitations and challenges, while Section 6 charts future directions. Section 7 concludes with practical recommendations for method selection.

2. Fundamentals of Metasurface Design

2.1. Generalized Snell's Law and Phase Gradient Metasurfaces

The operation of metasurfaces rests on the generalized laws of reflection and refraction derived by Yu *et al.* [2]. When a phase discontinuity $\Phi(x, y)$ is introduced at an interface between media of refractive indices n_i and n_t , Fermat's principle yields the generalized Snell's law (Eq. 1). This expression reveals that by engineering a spatially varying phase profile $\Phi(x, y)$ across the metasurface aperture, one can deflect, focus, or arbitrarily reshape an incident wavefront without relying on curvature or propagation distance. For example, a metalens that focuses a normally incident plane wave to a point at focal length f requires the hyperbolic phase profile

$$\Phi(x, y) = -\frac{2\pi}{\lambda_0} \left(\sqrt{x^2 + y^2 + f^2} - f \right), \quad (2)$$

which compensates for the path-length difference between the lens center and its periphery. The practical challenge is to realize this continuous phase profile using a discrete array of subwavelength scatterers, each providing a prescribed phase delay while maintaining high transmission amplitude.

2.2. Phase Control Mechanisms

2.2.1. Resonant Phase (Waveguide-Like)

High-index dielectric nanopillars—typically cylindrical or rectangular posts of TiO_2 , Si, or GaN on a substrate—act as truncated waveguides supporting Fabry–Pérot-like resonances along their vertical axis and Mie-type resonances determined by their lateral dimensions [3]. By varying the pillar diameter (or width and length for rectangular cross-sections) while keeping the height and period fixed, one can sweep the effective modal index and thereby control the transmitted phase over a full 2π range. This *resonant* or *propagation* phase mechanism offers the advantage of polarization-independent operation for circular pillars and can achieve high transmission

efficiencies ($> 90\%$) when material absorption is low. However, the phase response is intrinsically coupled to the transmission amplitude—near resonances, the amplitude dips—requiring careful design to maintain uniform amplitude across the library. The resonant phase is also inherently dispersive, with the phase–wavelength relationship governed by the pillar’s modal dispersion.

2.2.2. Geometric (Pancharatnam–Berry) Phase

An alternative phase control mechanism exploits the geometric or Pancharatnam–Berry (PB) phase, which arises from the orientation of anisotropic meta-atoms illuminated by circularly polarized light [4]. When a half-wave-plate-like meta-atom is rotated by an angle α about the optical axis, the cross-polarized transmitted component acquires a phase shift $\Phi = 2\alpha$, independent of the meta-atom’s spectral response. This mechanism is inherently *broadband*, since the 2α relationship is purely geometric and does not depend on wavelength. The trade-off is that it operates only on circularly polarized light, converting left-circular to right-circular (or vice versa), with the unconverted co-polarized component constituting a source of noise. Achieving high conversion efficiency requires the meta-atom to closely approximate a half-wave plate across the operating bandwidth, which imposes material and geometric constraints. Many state-of-the-art metasurfaces combine both resonant and geometric phases to achieve independent control of multiple polarization channels [15, 16].



Fig. 3. Comparison of phase control mechanisms in dielectric metasurfaces. (a) Resonant phase, where the phase delay is determined by the propagation constant of the waveguide mode governed by the nanopillar diameter D and height H . (b) Geometric (Pancharatnam–Berry) phase, where the phase delay depends solely on the orientation angle α of an anisotropic nanofin under circularly polarized illumination.

2.3. Material Platforms

The choice of material platform profoundly influences metasurface performance, fabrication complexity, and operating wavelength range. Table 1 summarizes the principal platforms. Titanium dioxide (TiO_2) has emerged as the material of choice for visible-wavelength metalenses due to its high refractive index ($n \approx 2.4$) and negligible absorption ($\kappa < 10^{-4}$) throughout the visible spectrum [5]. However, high-quality TiO_2 nanopillars require atomic layer deposition (ALD) combined with electron-beam lithography—a slow, serial process. Amorphous silicon (a-Si) offers higher index ($n \approx 3.5\text{--}4.0$) and is readily deposited by plasma-enhanced chemical vapor deposition (PECVD), but its significant absorption in the visible ($\kappa \sim 0.01\text{--}0.1$) restricts its use to near-infrared (NIR) wavelengths. Crystalline silicon (c-Si) on silicon-on-insulator (SOI) wafers is the workhorse for NIR to mid-infrared applications, benefiting from mature CMOS fabrication infrastructure. Gallium nitride (GaN) and silicon nitride (SiN) provide intermediate-index alternatives with visible transparency, while plasmonic metals (Ag, Au, Al)

137 remain relevant for applications requiring strong field confinement or operation in the ultraviolet,
 138 despite their inherently higher losses [4].

Table 1. Comparison of common material platforms for metasurface design.

Material	n (vis)	κ (vis)	Transparency	Fab. Method
TiO ₂	2.4	$< 10^{-4}$	Visible	ALD + EBL
a-Si	3.5–4.0	0.01–0.1	NIR	PECVD + EBL
c-Si	3.5	$< 10^{-4}$	NIR–MIR	SOI + EBL
GaN	2.4	$< 10^{-4}$	Visible	MOCVD + EBL
SiN	2.0	$< 10^{-5}$	Visible–NIR	LPCVD + EBL
Ag	—	—	Plasmonic	EBL/FIB
Au	—	—	Plasmonic	EBL/FIB

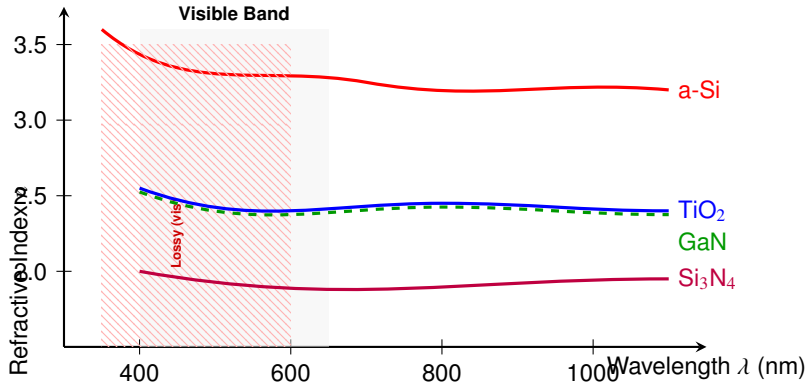


Fig. 4. Refractive index n and transparency regions for key metasurface material platforms across visible and near-infrared (NIR) wavelengths. The shaded visible band highlights the high losses (slashed region) of silicon compared to titanium dioxide, gallium nitride, and silicon nitride.

139 2.4. Unit Cell Design Principles

140 The standard approach to metasurface design begins with the construction of a *unit-cell library*:
 141 a comprehensive look-up table mapping geometric parameters (e.g., pillar diameter d , height h ,
 142 and period p) to the complex transmission coefficient $t(d) = |t|e^{i\Phi}$ at one or more wavelengths.
 143 This library is generated by simulating a single meta-atom under periodic boundary conditions—
 144 typically using FDTD (Section 3.1) or RCWA—and sweeping the relevant geometric parameters
 145 over their feasible range. A well-designed library provides full 2π phase coverage with near-
 146 uniform transmission amplitude ($|t| > 0.9$), ensuring that any target phase profile can be faithfully
 147 sampled.

148 Critically, this library-based approach relies on the *local phase approximation* (LPA): the
 149 assumption that each meta-atom's optical response is determined solely by its own geometry,
 150 independent of its neighbors. Under LPA, the phase and amplitude at each point on the metasurface

aperture are read directly from the library, treating the spatially varying device as a patchwork of locally periodic structures. The LPA is well justified when the phase profile varies slowly relative to the unit-cell period—i.e., when adjacent meta-atoms have similar geometries—but breaks down for rapidly varying profiles (high numerical aperture lenses, sharp phase gradients) or when strong near-field coupling between dissimilar neighbors significantly perturbs the optical response.

2.5. From Unit Cell to Device

Once the unit-cell library is established, the full metasurface device is assembled by assigning each lattice site the meta-atom geometry whose library-predicted phase best matches the target phase profile at that location. This process converts a continuous phase function $\Phi(x, y)$ into a discrete array of meta-atoms—a step that introduces quantization error proportional to the library’s phase sampling density and the spatial sampling period.

A fundamental issue arises: the optical response of each meta-atom in the assembled device may differ from its library-predicted response due to *near-field coupling* with neighboring elements. When adjacent meta-atoms have dissimilar geometries—as occurs near phase wraps, at the lens edge, or in high-NA regions—evanescent fields from one pillar can excite modes in its neighbors, altering both phase and amplitude. Full-wave simulation of the entire device would capture these effects, but is computationally prohibitive for devices larger than $\sim 50 \mu\text{m}$ (see Section 3.1). This gap between unit-cell-level design and device-level performance is one of the central challenges in metaoptics simulation, and motivates the multi-scale and hybrid approaches discussed in Section 3.4.

3. Simulation Methodologies

The simulation challenge in metaoptics is fundamentally *multi-scale*: individual meta-atoms have feature sizes of tens to hundreds of nanometers and must be modeled with sub-wavelength resolution, yet practical devices span millimeters to centimeters—a scale ratio of 10^4 to 10^5 . No single computational method efficiently bridges this range. Full-wave solvers provide the highest fidelity but are limited to small devices; propagation-based methods scale to macroscopic apertures but sacrifice accuracy at the unit-cell level; data-driven approaches offer speed but require careful validation. This section reviews four classes of methods—FDTD, Fourier optics, machine learning/inverse design, and hybrid multi-scale approaches—providing for each the mathematical foundation, practical implementation, metaoptics applications, and critical limitations.

3.1. Finite-Difference Time-Domain (FDTD)

3.1.1. Mathematical Foundation

The FDTD method solves Maxwell’s curl equations directly in the time domain by discretizing both space and time on a staggered grid introduced by Yee [17]. The electric and magnetic field components are interleaved in space (the *Yee cell*) and updated alternately in time via a leapfrog scheme. For a source-free region with permittivity ε and permeability μ , the update equations take the form

$$\mathbf{H}^{n+1/2} = \mathbf{H}^{n-1/2} - \frac{\Delta t}{\mu} \nabla \times \mathbf{E}^n, \quad (3)$$

$$\mathbf{E}^{n+1} = \mathbf{E}^n + \frac{\Delta t}{\varepsilon} \nabla \times \mathbf{H}^{n+1/2}, \quad (4)$$

190 where superscripts denote time steps and the spatial derivatives are approximated by second-order
 191 central differences. Numerical stability requires the Courant–Friedrichs–Lewy (CFL) condition

$$c \Delta t \leq \left(\sum_{i=1}^d \frac{1}{\Delta x_i^2} \right)^{-1/2}, \quad (5)$$

192 where c is the speed of light, d is the dimensionality, and Δx_i are the grid spacings. For
 193 metaoptics simulations at visible wavelengths ($\lambda \sim 400\text{--}700$ nm), grid spacings of $\Delta x \leq \lambda/(20n)$
 194 are typically required, where n is the highest refractive index in the simulation domain, to ensure
 195 convergence. This translates to grid cells of $\sim 5\text{--}15$ nm for high-index dielectric metasurfaces.

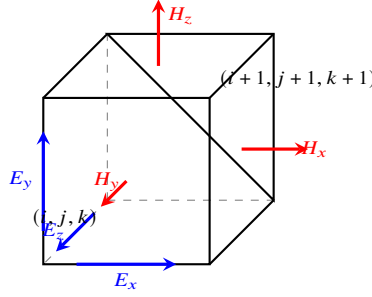


Fig. 5. Schematic of the Yee grid discretization cell used in 3D FDTD simulations. Electric field components (E_x, E_y, E_z , blue arrows) are placed on the cell edges, while magnetic field components (H_x, H_y, H_z , red arrows) are centered on the cell faces, staggered by half a grid cell in both space and time.

196 3.1.2. Boundary Conditions

197 The choice of boundary conditions critically determines both the physical fidelity and computa-
 198 tional cost of FDTD simulations. For unit-cell library generation, *periodic boundary conditions*
 199 (PBCs) model an infinite array of identical meta-atoms, capturing the optical response under
 200 the assumption of local periodicity. For oblique incidence, Bloch-periodic boundaries with an
 201 appropriate phase shift across the unit cell are required. The simulation domain is truncated in the
 202 propagation direction using *perfectly matched layers* (PMLs)—artificial absorbing regions that
 203 attenuate outgoing waves with minimal reflection, typically achieving reflectivities below 10^{-6} .
 204 For finite-size device simulations, PMLs surround the entire computational domain. Symmetric
 205 and antisymmetric boundary conditions can exploit device symmetries to reduce computational
 206 cost by factors of 2–8, a significant advantage for 3D simulations.

207 3.1.3. Material Models

208 Accurate representation of frequency-dependent material properties is essential for broadband
 209 metaoptics simulation. In FDTD, dispersive materials are modeled through auxiliary differential
 210 equations (ADEs) that couple the field update equations to polarization currents. The Drude
 211 model captures the free-electron response of metals:

$$\varepsilon(\omega) = \varepsilon_\infty - \frac{\omega_p^2}{\omega^2 + i\gamma\omega}, \quad (6)$$

212 where ω_p is the plasma frequency and γ the damping rate. Dielectric resonances are described
 213 by Lorentz oscillators, and general dispersive media are fitted using multi-coefficient models with

multiple Drude and Lorentz poles. Modern FDTD software automates this fitting process, but achieving broadband accuracy—particularly for lossy, anisotropic, or phase-change materials—remains a non-trivial challenge that directly impacts simulation reliability.

3.1.4. Near-to-Far-Field Transformation

FDTD simulations compute fields on a finite near-field grid, but metaoptics design requires knowledge of the *far-field* pattern—the focal spot of a metalens, the diffraction pattern of a hologram, or the radiation diagram of a beam shaper. The near-to-far-field transformation (NFTF), based on the surface equivalence theorem, extracts equivalent electric and magnetic surface currents on a closed surface surrounding the device and propagates them analytically to arbitrary observation points in the far field. This transformation is computationally inexpensive compared to extending the FDTD grid to the focal plane and is a standard feature of all major FDTD packages. However, for metaoptics applications, the NFTF is often used in conjunction with Fourier propagation techniques (Section 3.2) for more flexible analysis of focal properties, aberrations, and imaging performance.

3.1.5. Software Ecosystem

The metaoptics community relies on both commercial and open-source FDTD implementations (Table 2). *Lumerical FDTD Solutions* (now part of Ansys) is the most widely used commercial package, offering a mature graphical interface, Python/Lua scripting, GPU acceleration, and built-in parameter sweep and optimization capabilities. *MEEP* (MIT Electromagnetic Equation Propagation) [18] is the leading open-source alternative, supporting MPI-based distributed computing for large-scale simulations on high-performance computing (HPC) clusters. *CST Studio Suite* provides a multi-solver environment that includes FDTD alongside frequency-domain and integral-equation solvers. More recently, *Tidy3D* (Flexcompute) has emerged as a cloud-based, GPU-accelerated FDTD service claiming speedups of 100× or more over CPU-based implementations, potentially shifting the scalability boundary for full-wave metaoptics simulation.

Table 2. Comparison of FDTD software packages for metaoptics simulation.

Software	License	GPU	Scripting	Typical Use
Lumerical	Commercial	Yes	Python/Lua	Unit cell, small device
MEEP	Free (GPL)	No	Python/Scheme	Research, HPC
CST Studio	Commercial	Yes	VBA/Python	Broadband, antenna

3.1.6. Applications in Metaoptics

FDTD is the gold standard for metaoptics simulation in three principal contexts. First, it is the primary tool for *unit-cell library generation*: parametric sweeps over nanopillar geometry (diameter, height, aspect ratio) under periodic boundary conditions yield the phase–amplitude look-up tables that underpin the library-based design approach (Section 2.4). A typical library sweep involves 10^2 – 10^3 individual 3D simulations, each requiring minutes to hours depending on resolution and material complexity. Second, FDTD enables *full-wave simulation of small-aperture devices* ($< 50 \mu\text{m}$ diameter), providing rigorous validation that captures near-field coupling, edge effects, and diffraction not accounted for by the LPA. Such simulations have been instrumental in understanding performance degradation mechanisms in high-NA metalenses. Third, FDTD is indispensable for *resonance engineering*—designing meta-atoms that support

sharp spectral features such as bound states in the continuum (BICs) [19], Fano resonances, or guided-mode resonances, where the precise modal structure dictates device performance.

3.1.7. Limitations

The fundamental limitation of FDTD for metaoptics is its *computational cost*, which scales as $O(N^d)$ in memory and $O(N^{d+1})$ in time, where N is the number of grid points per dimension and d is the spatial dimensionality. For a 3D simulation of a metalens with diameter D operating at wavelength λ in a material of index n , the required number of grid cells is approximately $(20nD/\lambda)^2 \times (20nH/\lambda)$, where H is the domain height. A modest 100 μm -diameter metalens at $\lambda = 500$ nm in TiO_2 ($n = 2.4$) with $H = 2$ μm requires $\sim (10^4)^2 \times 200 \approx 2 \times 10^{10}$ cells, corresponding to ~ 150 GB of memory—already challenging for current workstations. A 1-mm lens would require ~ 15 TB, and a centimeter-scale device would exceed any foreseeable computing resource. Additional limitations include the *staircase approximation*—curved and angled geometries represented on a Cartesian grid suffer from discretization error unless subpixel averaging or conformal meshing is employed—and the difficulty of handling *broadband excitation* for highly dispersive structures, where very long simulation times are needed to resolve narrow spectral features.

3.2. Ray Tracing and Fourier Optics

3.2.1. Scalar Diffraction Theory

Whereas FDTD solves the full vectorial Maxwell's equations, many metaoptics design tasks can be addressed more efficiently using *scalar diffraction theory*, which describes light propagation as a scalar field $U(x, y, z)$ satisfying the Helmholtz equation [20]. The Huygens–Fresnel principle treats each point on a wavefront as a secondary source of spherical wavelets; their superposition at an observation plane yields the Rayleigh–Sommerfeld diffraction integral:

$$U(x, y, z) = \frac{1}{i\lambda} \iint U(x', y', 0) \frac{e^{ikr}}{r} \cos \theta \, dx' \, dy', \quad (7)$$

where $r = \sqrt{(x - x')^2 + (y - y')^2 + z^2}$ and θ is the angle between the propagation direction and the surface normal. Under the paraxial approximation ($z \gg |x - x'|, |y - y'|$), this simplifies to the Fresnel diffraction integral, and in the far field ($z \gg D^2/\lambda$, where D is the aperture diameter) to the Fraunhofer integral—a simple Fourier transform. For metaoptics, the Fresnel regime is most relevant, as focal lengths are typically comparable to or smaller than the Fraunhofer transition distance.

3.2.2. Angular Spectrum Method (ASM)

The angular spectrum method (ASM) provides a numerically exact (within the scalar approximation) and computationally efficient framework for propagating optical fields over arbitrary distances. The field at the metasurface plane $U(x, y, 0)$ is decomposed into a spectrum of plane waves via a 2D Fourier transform:

$$\tilde{U}(k_x, k_y, 0) = \mathcal{F}\{U(x, y, 0)\}, \quad (8)$$

where k_x, k_y are the transverse spatial frequencies. Each plane-wave component propagates independently, acquiring a phase factor $e^{ik_z z}$ with $k_z = \sqrt{k^2 - k_x^2 - k_y^2}$. The field at distance z is then obtained by an inverse Fourier transform:

$$U(x, y, z) = \mathcal{F}^{-1}\{\tilde{U}(k_x, k_y, 0) \cdot e^{ik_z z}\}. \quad (9)$$

287 This amounts to two fast Fourier transforms (FFTs) and a pointwise multiplication, making
 288 ASM extremely efficient for large-aperture devices—a centimeter-scale metalens sampled at
 289 $\lambda/2$ requires a $\sim 10^4 \times 10^4$ FFT, which executes in seconds on a modern workstation. Careful
 290 attention to sampling is required: the spatial sampling must satisfy the Nyquist criterion for
 291 the highest spatial frequency present (set by the finest feature or steepest phase gradient), and
 292 zero-padding or band-limiting may be needed to suppress aliasing artifacts at large propagation
 293 distances.

294 3.2.3. Geometric Ray Tracing

295 At device scales much larger than the wavelength, geometric (ray) optics provides a complementary
 296 perspective. In this framework, the metasurface is modeled as a *thin phase element*: an
 297 infinitesimally thin surface that imparts a spatially varying phase $\Phi(x, y)$ to transmitted rays
 298 without altering their position. The local deflection angle is determined by the phase gradient
 299 via Eq. 1. This representation allows metasurfaces to be integrated directly into commercial
 300 ray-tracing software (e.g., Zemax OpticStudio, Code V) alongside conventional refractive and
 301 reflective elements, enabling the design of multi-element metaoptical systems, evaluation of
 302 field curvature and distortion, and optimization of imaging performance metrics such as the
 303 modulation transfer function (MTF). However, ray tracing inherently neglects diffraction effects
 304 and is therefore unable to predict focal spot size, diffraction efficiency, or performance near the
 305 diffraction limit.

306 3.2.4. The Standard Hybrid Workflow

307 The *de facto* standard simulation workflow in metaoptics combines FDTD at the unit-cell scale
 308 with Fourier propagation at the device scale, bridging the multi-scale gap in a pragmatic manner.
 309 The workflow proceeds in three steps: (1) *Library generation*: FDTD or RCWA simulations of
 310 individual meta-atoms under periodic boundary conditions produce a look-up table mapping
 311 geometric parameters to complex transmission coefficients $t_j = |t_j|e^{i\Phi_j}$. (2) *Spatial mapping*:
 312 The target phase profile $\Phi(x, y)$ (e.g., Eq. 2) is discretized on the device aperture, and each lattice
 313 site is assigned the meta-atom geometry from the library whose phase best matches the target.
 314 The assembled complex transmission profile $t(x, y)$ serves as the input field for propagation.
 315 (3) *Fourier propagation*: The transmitted field $U_{\text{out}}(x, y) = U_{\text{in}}(x, y) \cdot t(x, y)$ is propagated to
 316 the focal plane using the ASM, yielding the point spread function (PSF), focusing efficiency,
 317 and other performance metrics. This workflow is computationally efficient (the ASM step takes
 318 seconds even for centimeter-scale devices) and has been used in the vast majority of metalens
 319 demonstrations reported in the literature [5, 6, 21].

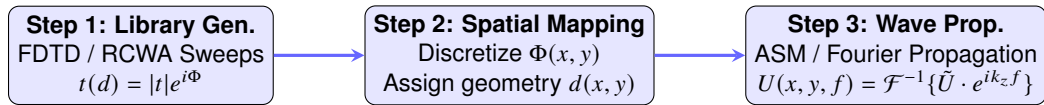


Fig. 6. The standard hybrid simulation workflow in metaoptics. Step 1: Full-wave unit-cell library generation mapping pillar dimensions to transmission phase and amplitude. Step 2: Discretizing the target phase profile and mapping geometries onto the array. Step 3: Fast Fourier / Angular Spectrum propagation to evaluate device performance.

320 3.2.5. Applications in Metaoptics

321 Fourier-optics-based methods are the workhorse for performance prediction of *large-scale*
 322 metaoptical devices. They enable evaluation of the point spread function (PSF) and modulation
 323 transfer function (MTF) of metalenses with apertures from hundreds of micrometers to

centimeters—device sizes completely inaccessible to full-wave methods. Chromatic aberration analysis benefits particularly, as ASM propagation can be repeated at multiple wavelengths with different unit-cell library entries to map the focal shift, Strehl ratio, and PSF broadening across the operating bandwidth. For achromatic metalens design (Section 4.4), this multi-wavelength propagation is performed iteratively during the optimization loop. Ray tracing extends the analysis to *multi-element metaoptical systems*—for example, a metalens doublet corrected for chromatic and geometric aberrations—where system-level optimization requires thousands of field evaluations that are feasible only with ray-based or Fourier methods [21].

3.2.6. Limitations

The fundamental limitation of Fourier-optics-based methods is their reliance on the *local phase approximation* (LPA), which treats each meta-atom as an independent phase-shifting element and ignores electromagnetic coupling between neighbors. This approximation is well justified for slowly varying phase profiles but introduces systematic errors for high-NA metalenses ($NA > 0.7$), where adjacent meta-atoms may differ significantly in geometry and evanescent-field coupling becomes pronounced. Several studies have quantified this breakdown, showing that LPA-predicted focal efficiencies can overestimate full-wave results by 10–30% for high-NA designs. Additionally, scalar diffraction theory neglects *vectorial* effects (polarization-dependent focusing, depolarization at high NA) and *multiple scattering* (back-reflections between the metasurface and substrate or detector), effects that may be significant in practical systems. These limitations motivate the hybrid and multi-scale approaches discussed in Section 3.4.

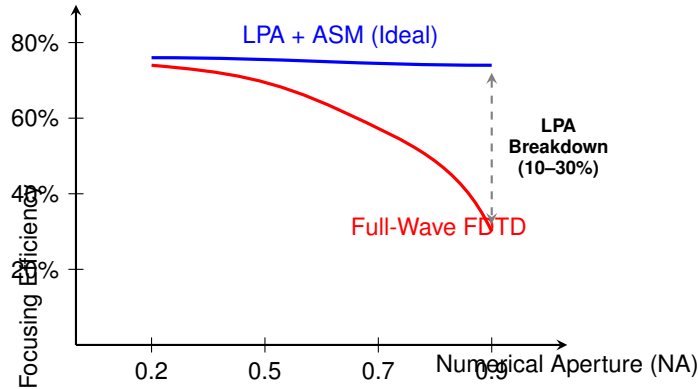


Fig. 7. Comparison of focusing efficiency vs. Numerical Aperture (NA) predicted by the Local Phase Approximation (LPA) combined with ASM (blue line) versus full-wave 3D FDTD simulations (red line). As NA increases beyond 0.7, the rapid spatial variation of the phase profile causes strong near-field coupling, violating the LPA and leading to a significant drop in actual full-wave efficiency.

3.3. Machine Learning and Inverse Design

3.3.1. Forward vs. Inverse Problem

The design of metaoptical devices can be framed as either a *forward* or an *inverse* problem. The forward problem—given a meta-atom geometry, predict the optical response (transmission amplitude, phase, spectrum)—is well-posed and solved by electromagnetic solvers such as FDTD or RCWA. The inverse problem—given a desired optical response, determine the geometry that produces it—is fundamentally *ill-posed*: multiple geometries may yield identical or near-identical responses (non-uniqueness), and small perturbations in the target can lead to large changes in

the optimal geometry (instability). Traditional design approaches sidestep the inverse problem through exhaustive parametric sweeps or heuristic optimization (genetic algorithms, particle swarm), both of which scale poorly to high-dimensional design spaces. Machine learning and inverse design methods aim to address this challenge by either learning the forward or inverse mapping from data, or by leveraging gradient-based optimization to navigate the design space efficiently [12, 22].

3.3.2. Neural Network Surrogates

The most straightforward application of deep learning to metaoptics is the *forward surrogate*: a neural network trained on FDTD- or RCWA-generated datasets to predict the optical response of a meta-atom from its geometric parameters [12, 13]. Once trained, a forward surrogate executes in milliseconds per evaluation—orders of magnitude faster than a full-wave simulation—enabling rapid exploration of the design space, real-time optimization, and integration into iterative design loops. The training dataset is typically generated by simulating 10^3 – 10^5 randomly or systematically sampled geometries, and the network architecture ranges from simple multilayer perceptrons (MLPs) to convolutional neural networks (CNNs) for spatially resolved designs.

Inverse networks directly predict the geometry from a target response, but suffer from the non-uniqueness problem: multiple geometries map to the same spectrum, causing the network to predict an unphysical average. The *tandem network* architecture [14] resolves this by coupling a pre-trained forward network to an inverse network, using the forward network as a differentiable physics proxy during backpropagation. This ensures that the predicted geometry actually produces the desired response, even if other valid solutions exist.

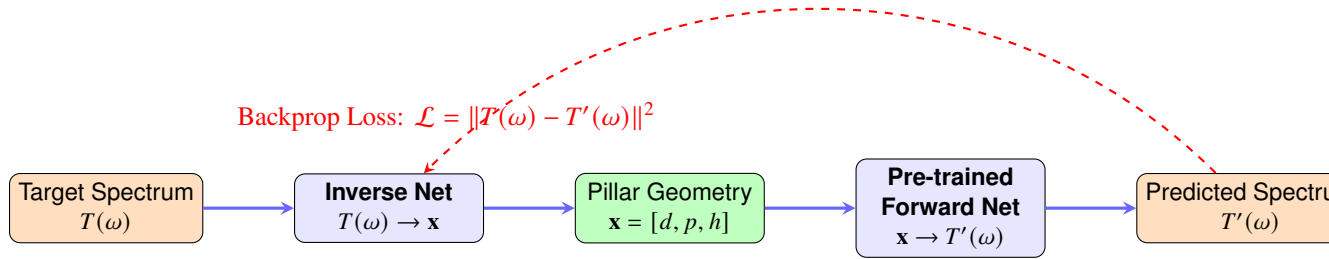


Fig. 8. The Tandem Neural Network architecture for solving the ill-posed metasurface inverse design problem. A pre-trained, frozen forward surrogate network acts as a differentiable physics proxy. During backpropagation, the loss between the target spectrum $T(\omega)$ and the predicted spectrum $T'(\omega)$ is backpropagated through the forward network to update the weights of the inverse network, preventing the non-uniqueness problem from stalling convergence.

3.3.3. Generative Models

Beyond regression-based surrogates, *generative models* can create entirely new meta-atom topologies not present in the training data. Generative adversarial networks (GANs) learn to produce realistic unit-cell geometries that satisfy target optical specifications, while variational autoencoders (VAEs) provide a continuous latent space that enables smooth interpolation between designs and targeted exploration of novel topologies. Conditional variants of both architectures accept the target spectrum or phase as input and generate geometries that achieve it. More recently, *diffusion models*—which have revolutionized image generation—are being adapted for metasurface design, offering superior mode coverage and training stability compared to GANs.

382 These generative approaches are particularly valuable for discovering *freeform* meta-atom shapes
383 (e.g., irregularly shaped holes or connected patterns) that would not be explored by traditional
384 parametric sweeps.

385 3.3.4. Topology Optimization and Adjoint Method

386 Topology optimization (TO) takes a fundamentally different approach: rather than learning
387 from data, it directly optimizes the spatial distribution of material within a design region to
388 maximize a specified objective function [22–24]. The key enabling technique is the *adjoint*
389 *method*, which computes the gradient of the objective with respect to *all* design parameters (e.g.,
390 the permittivity at every pixel) using just *two* electromagnetic simulations—one forward and
391 one adjoint—regardless of the number of parameters. This makes gradient-based optimization
392 tractable even for designs with 10^4 – 10^6 degrees of freedom.

393 In density-based TO, the design region is discretized into pixels whose permittivity varies
394 continuously between air and the target material (e.g., Si). The optimization converges to a
395 binary (0/1) material distribution through binarization penalties or projection filters. Level-set
396 methods represent material boundaries as the zero contour of a smooth function, providing
397 sharper interfaces. Both approaches can incorporate *fabrication constraints*—minimum feature
398 size, minimum spacing, connectivity requirements, maximum aspect ratio—as additional
399 regularization terms or geometric filters, producing designs that are manufacturable by standard
400 lithographic processes. The resulting freeform geometries often outperform intuitive designs
401 (nanopillars, nanofins) by exploiting non-obvious modal interactions [23, 25].

402 3.3.5. Physics-Informed Approaches

403 Physics-informed neural networks (PINNs) embed the governing physical equations—Maxwell’s
404 equations, constitutive relations—directly into the loss function, constraining the network to
405 produce physically consistent predictions even with limited training data [10]. Unlike purely
406 data-driven surrogates, PINNs do not require large FDTD-generated datasets; instead, they learn
407 to satisfy the differential equations at collocation points sampled throughout the computational
408 domain. This hybrid data-physics approach offers improved generalization and reduced training
409 cost, at the expense of more complex loss landscapes and potentially slower convergence. PINNs
410 are particularly promising for problems where training data is expensive to generate (e.g., 3D
411 broadband simulations) or where extrapolation beyond the training distribution is required.

412 3.3.6. Applications in Metaoptics

413 ML and inverse design methods have been applied across the full spectrum of metaoptics appli-
414 cations (Table 3). Forward surrogates accelerate unit-cell optimization by 10^3 – $10^6\times$ compared
415 to FDTD, enabling real-time design exploration and high-throughput screening of material–
416 geometry combinations. Topology optimization has produced record-efficiency metasurfaces [23]
417 and large-scale meta-optics for virtual reality [25]. Multi-objective optimization—simultaneously
418 targeting high efficiency, broadband operation, and polarization control—is naturally handled by
419 gradient-based methods and has yielded multifunctional devices that would be difficult to design
420 by intuition alone. Perhaps most transformatively, ML-based surrogates have been integrated
421 into *end-to-end differentiable pipelines* that jointly optimize the physical metasurface and the
422 computational post-processing algorithm (e.g., a neural network image reconstructor), blurring
423 the boundary between optics design and computational imaging [10].

424 3.3.7. Limitations

425 Despite their promise, ML and inverse design methods face significant limitations. *Data*
426 *dependency* is paramount: neural network surrogates are only as good as their training data,
427 and predictions for geometries, materials, or wavelength ranges outside the training distribution

Table 3. Comparison of machine learning and inverse design approaches for metaoptics.

Approach	Training Cost	Inference Speed	Generalization	Fab. Constraints
Forward NN	High	ms	Moderate	Post-hoc
Inverse NN	High	ms	Low	Post-hoc
Tandem NN	High	ms	Moderate	Post-hoc
GAN/VAE	Very High	ms	Low	Implicit
Adjoint/TO	N/A (per-design)	Hours	High	Built-in
PINN	Moderate	Seconds	High	Built-in

are unreliable. Generating sufficient training data—typically 10^4 – 10^5 FDTD simulations—represents a substantial upfront computational investment. *Generalization* remains challenging: a surrogate trained on TiO_2 nanopillars at visible wavelengths cannot be directly applied to Si nanofins at near-infrared wavelengths without retraining. *Black-box* designs produced by topology optimization or generative models often lack physical interpretability, making it difficult to diagnose failure modes or extract design rules. Critically, *all ML-predicted designs must be validated by full-wave simulation* before fabrication, as surrogate errors can compound through the design pipeline. Finally, the encoding of *fabrication constraints*—minimum feature size, sidewall angle, etch selectivity—into ML and TO frameworks remains an active area of research, with no universally accepted methodology [12, 13].

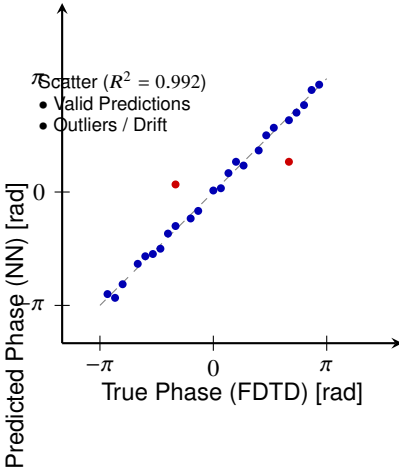


Fig. 9. A representative prediction accuracy plot of a forward neural network surrogate compared against full-wave FDTD simulations. The high R^2 value and tight clustering along the $y = x$ diagonal line indicate excellent model accuracy for in-distribution geometries, while rare outliers (red dots) demonstrate the risk of out-of-distribution drift.

438 3.4. Combined and Multi-Scale Methods

439 3.4.1. Multi-Scale Simulation Strategies

440 The most widely adopted multi-scale strategy is the FDTD–Fourier hybrid described in Section 3.2,
441 which pairs rigorous unit-cell simulation with efficient scalar propagation. Variants of this
442 approach replace the FDTD unit-cell stage with faster alternatives: *RCWA* (rigorous coupled-wave
443 analysis) offers frequency-domain solutions that are faster than FDTD for periodic structures,
444 while *ML surrogates* (Section 3.3) can replace electromagnetic solvers entirely during iterative
445 optimization, reducing the unit-cell evaluation cost by orders of magnitude. At the system level,
446 *FDTD + ray tracing* combinations model the metasurface rigorously while treating bulk optical
447 elements (lenses, mirrors) geometrically, enabling the design of hybrid refractive–diffractive
448 systems. *Domain decomposition* methods partition a large metasurface into smaller overlapping
449 regions, solve each with FDTD, and stitch the solutions together, partially capturing inter-cell
450 coupling at reduced cost.

451 3.4.2. Transfer Matrix and Coupled-Mode Theory

452 Analytical and semi-analytical methods complement numerical simulations by providing physical
453 insight and rapid parameter-space exploration. The *transfer matrix method* (TMM) models
454 multilayer metasurface stacks—for example, a cascaded pair of metasurfaces separated by a
455 spacer—by representing each layer as a 2×2 (scalar) or 4×4 (vectorial) matrix that relates forward-
456 and backward-propagating fields. TMM is exact for planar stratified media and computationally
457 trivial, making it ideal for optimizing spacer thickness, substrate effects, and anti-reflection
458 coatings. *Temporal coupled-mode theory* (TCMT) describes the interaction of incident light
459 with resonant modes of meta-atoms through a small number of parameters (resonant frequency,
460 radiative and non-radiative decay rates, coupling coefficients), providing closed-form expressions
461 for transmission and reflection spectra near resonance. TCMT is particularly valuable for
462 designing high- Q sensing metasurfaces (Section 4.6) and understanding Fano and BIC resonance
463 physics.

464 3.4.3. End-to-End Differentiable Pipelines

465 The most transformative recent development in computational metaoptics is the emergence
466 of *end-to-end differentiable* design pipelines that connect all stages—unit-cell response, array
467 assembly, wave propagation, detector response, and computational reconstruction—in a single,
468 automatically differentiable computational graph [10]. Implemented in frameworks such as
469 PyTorch or JAX, these pipelines enable gradient-based optimization of the *entire* imaging or
470 sensing system, including both the physical metasurface and the downstream computational
471 algorithm. For example, a metalens can be jointly optimized with a neural network deconvolution
472 algorithm: the lens is allowed to exhibit controlled aberrations that are computationally corrected,
473 resulting in a system whose combined optical–computational performance exceeds that of a
474 conventionally optimized lens alone. This *computational metaoptics* paradigm (Section 4.7)
475 represents a fundamental shift from designing optics for perfect image formation to designing
476 optics for optimal information capture.

477 3.4.4. Limitations

478 Hybrid and multi-scale methods inherit the limitations of their constituent components and
479 introduce additional challenges at the interfaces between scales. *Error accumulation* is a primary
480 concern: inaccuracies in the unit-cell simulation (e.g., neglected substrate effects, truncated
481 simulation time) propagate through the Fourier propagation stage and can compound to produce
482 significant discrepancies at the device level. The *interface* between different simulation domains
483 requires careful matching of field representations, coordinate systems, and sampling grids;

Table 4. Comprehensive comparison of simulation methods for metaoptics. “Device scale” indicates the maximum device size practically simulatable.

Method	Accuracy	Device Scale	Speed	Memory	Coupling	Broadband	Open-Source
FDTD	Highest	$\sim 50 \mu\text{m}$	Slow	Very High	Full	Native	MEEP
RCWA	High	Unit cell	Moderate	Low	Periodic only	Per-freq.	S4, Reticolo
Fourier/ASM	Moderate	cm+	Fast	Low	None (LPA)	Native	Custom
Ray Tracing	Low–Mod.	cm+	Fast	Minimal	None	Geometric	Zemax
ML Surrogate	Variable	cm+ (w/prop.)	Very Fast	Low	Learned	Trained range	Various
Adjoint/TO	High	$\sim 100 \mu\text{m}$	Slow	High	Full	Per-freq.	Various
Hybrid	High	cm+	Moderate	Moderate	Partial	Depends	Custom

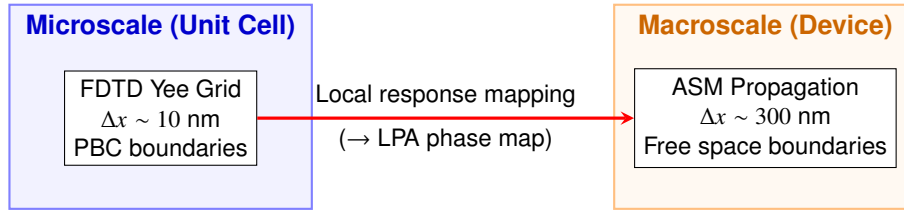


Fig. 10. Schematic of a multi-scale simulation pipeline, showing how microscale electromagnetics (unit-cell FDTD with periodic boundaries on a nanometer grid) couples to macroscale device evaluation (Angular Spectrum Method on a wavelength-scale grid) through Local Phase Approximation mapping.

484 mismatches can introduce artifacts such as spurious diffraction or energy non-conservation.
 485 Perhaps most critically, the field currently *lacks standardized multi-scale frameworks*: most
 486 research groups implement bespoke pipelines, making it difficult to compare results across
 487 studies or to reproduce published designs. The development of open-source, validated, and
 488 benchmarked multi-scale simulation tools represents one of the most pressing needs in the
 489 metaoptics community.

490 4. Application Domains

491 Each metaoptics application domain imposes distinct requirements on simulation fidelity,
 492 computational cost, and the physical effects that must be captured. In this section, we survey
 493 seven key application areas through the lens of which computational methods are employed and
 494 why, highlighting the simulation challenges specific to each domain.

495 4.1. Metalens Design and Optimization

496 Metalens design exemplifies the multi-scale simulation challenge. The target hyperbolic phase
 497 profile (Eq. 2) must be discretized into a unit-cell array, with each meta-atom selected from a
 498 phase library generated by FDTD or RCWA under periodic boundary conditions. The assembled
 499 device is then evaluated using ASM propagation to compute the point spread function (PSF),
 500 Strehl ratio, and focusing efficiency [5, 21].

501 For *imaging* metalenses, the analysis extends beyond the on-axis PSF to include off-axis
 502 aberrations (coma, astigmatism, field curvature), which are evaluated by propagating oblique-

503 incidence fields or by interfacing the metalens phase profile with ray-tracing software. Achromatic
 504 metalens design (Section 4.4) requires multi-wavelength library generation and chromatic PSF
 505 analysis, multiplying the computational burden. *High-NA* metalenses ($NA > 0.7$) present the
 506 greatest simulation challenge: the LPA breaks down as adjacent meta-atoms diverge in geometry
 507 near the lens edge, and vectorial diffraction effects become significant [6]. Full-wave simulation
 508 would capture these effects but is limited to apertures of $\sim 50\ \mu\text{m}$, while practical imaging
 509 metalenses require millimeter-to-centimeter apertures—a gap that remains the most pressing
 510 open problem in metalens simulation.

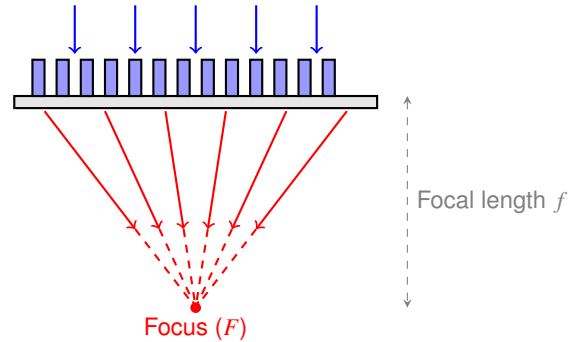


Fig. 11. Schematic of a flat metalens focusing collimated incident light (blue rays) to a single diffraction-limited spot at a focal distance f (red rays) through spatially varying subwavelength phase shifts.

511 4.2. Beam Shaping and Wavefront Control

512 Metasurfaces excel at generating structured light fields, including vortex beams carrying orbital
 513 angular momentum (OAM), non-diffracting Bessel and Airy beams, and arbitrary vector beams
 514 with spatially varying polarization [15,26]. Beam-shaping metasurfaces are typically designed
 515 using geometric (PB) phase elements, where the rotation angle of each meta-atom directly
 516 encodes the target phase, or using combined resonant–geometric phase for independent control
 517 of multiple polarization channels [16,27].

518 The simulation workflow is predominantly *phase-only design* validated by Fourier optics: the
 519 target complex field is decomposed into phase and amplitude components, the metasurface is
 520 designed to produce the required phase (and optionally amplitude), and ASM propagation verifies
 521 the output beam profile. The key simulation challenge is ensuring high *mode purity*—the fraction
 522 of total power in the desired mode—which requires accurate modeling of both the phase and
 523 amplitude response of each meta-atom. For OAM beams with high topological charge, the rapid
 524 azimuthal phase variation can stress the LPA and degrade mode purity.

525 4.3. Metasurface Holography

526 Metasurface holography encodes a target image into the phase and/or amplitude profile of a
 527 metasurface, which reconstructs the image upon illumination [28]. The design process typically
 528 involves iterative phase-retrieval algorithms—variants of the Gerchberg–Saxton (GS) algorithm
 529 that alternate between the metasurface and image planes, imposing constraints in each—to
 530 determine the optimal phase distribution. The computed phase is then discretized using a unit-cell
 531 library.

532 Advanced holographic metasurfaces exploit *multiplexing* to encode multiple images accessible
 533 through different wavelengths, polarization states, incidence angles, or OAM channels, enabled
 534 by multi-parameter meta-atom libraries. From a simulation perspective, holographic design

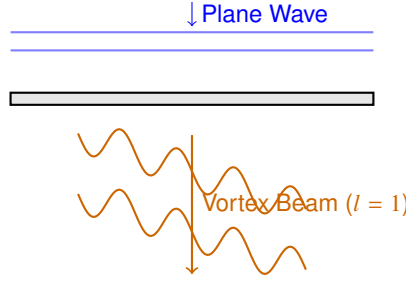


Fig. 12. Schematic of wavefront shaping, where a flat metasurface converts an incident plane wave (blue) into a helical wavefront (orange) possessing orbital angular momentum (OAM) with topological charge $l = 1$.

relies heavily on *iterative Fourier methods*—thousands of FFT-based forward and backward propagations per design iteration—making computational speed critical. FDTD enters only at the unit-cell library stage and for final verification of the designed hologram. The primary simulation limitation is the discrepancy between the idealized GS-designed phase profile and the quantized, amplitude-modulated profile realized by the actual meta-atom library.

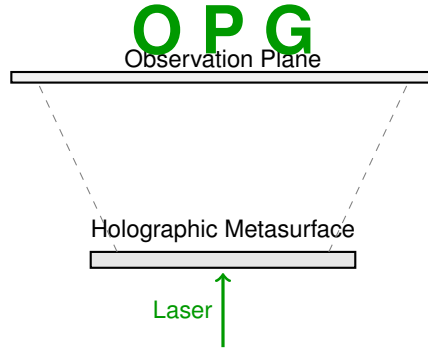


Fig. 13. Schematic of metasurface holography, showing reconstruction of a pre-designed multi-letter image (“OPG”) in the far-field observation plane upon laser illumination.

4.4. Broadband and Achromatic Metaoptics

Chromatic dispersion—the wavelength-dependent focal shift inherent to diffractive optics—is one of the most fundamental challenges in metaoptics. Unlike refractive lenses, where dispersion arises from material properties and can be corrected by combining positive and negative elements, diffractive dispersion is structural and much stronger [29, 30].

Achromatic metalens design requires meta-atoms that simultaneously provide the correct phase at multiple wavelengths while also matching the required *group delay* ($GD = d\Phi/d\omega$) and *group delay dispersion* ($GDD = d^2\Phi/d\omega^2$). This demands a *multi-wavelength* unit-cell library in which each meta-atom is characterized by its full spectral phase response $\Phi(\omega)$, typically computed via broadband FDTD simulations. The design optimization then searches for meta-atom sequences that jointly satisfy the phase, GD, and GDD targets at each spatial location—a combinatorial problem that benefits enormously from ML-accelerated surrogate models [25]. The key simulation challenge is that broadband FDTD is computationally expensive (long simulation times to resolve spectral features), and achieving achromaticity over the full

554 visible band ($\lambda = 400\text{--}700\text{ nm}$) with high efficiency remains an unsolved problem at macroscopic
 555 apertures.

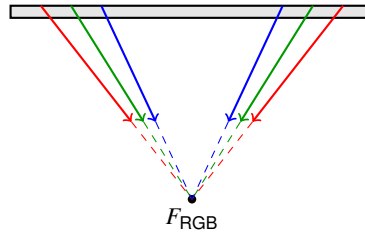


Fig. 14. Schematic of a broadband achromatic metalens, demonstrating simultaneous focusing of red, green, and blue incident wavelengths to a common spatial focal point F_{RGB} via dispersion engineering.

556 4.5. Tunable and Reconfigurable Metasurfaces

557 Tunable metasurfaces whose optical response can be dynamically reconfigured after fabrication
 558 represent one of the most active research frontiers [31–33]. Tuning mechanisms include: (i)
 559 *phase-change materials* (PCMs) such as $\text{Ge}_2\text{Sb}_2\text{Te}_5$ (GST), VO_2 , and Sb_2Se_3 , which undergo
 560 reversible crystalline-to-amorphous or metal-to-insulator transitions; (ii) *liquid crystals* infiltrated
 561 into or overlaid on meta-atom arrays, tuned by voltage-controlled birefringence; (iii) *MEMS*
 562 actuators that mechanically displace metasurface layers; and (iv) *electro-optical* modulation in
 563 materials such as ITO or graphene.

564 Simulation of tunable metasurfaces poses unique challenges because the optical response
 565 depends not only on the electromagnetic problem but also on *thermal* (PCM switching, Joule
 566 heating), *mechanical* (MEMS deflection, stress), or *electrochemical* (voltage-driven index change)
 567 physics. *Multi-physics simulation*—coupling FDTD with finite-element thermal or mechanical
 568 solvers—is required but significantly increases computational cost and complexity. Dynamic
 569 simulation frameworks that capture the *temporal evolution* of the metasurface state during
 570 switching are still in their infancy.

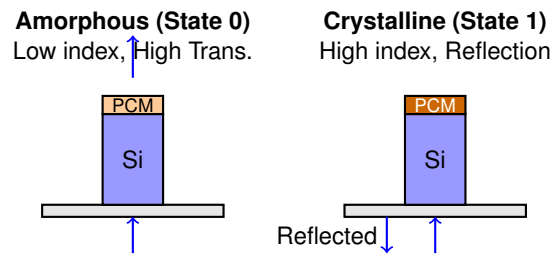


Fig. 15. Schematic of a tunable meta-atom utilizing a phase-change material (PCM) cladding. By switching the PCM from its amorphous state (State 0, low index) to its crystalline state (State 1, high index), the transmission and phase characteristics of the meta-atom are dynamically reconfigured.

571 4.6. Metaoptics for Sensing and Spectroscopy

572 Metasurfaces supporting high-quality-factor (Q) resonances are increasingly employed as
 573 sensitive optical sensors for detecting refractive index changes, molecular binding, and chiral
 574 analytes [34, 35]. The key simulation requirement is the accurate prediction of resonant line

575 shapes and Q -factors, which demands fine spectral and spatial resolution in full-wave simulations.
 576 *Bound states in the continuum* (BICs)—modes that are completely decoupled from the radiation
 577 continuum by symmetry—provide theoretically infinite Q -factors in ideal infinite arrays [19]; in
 578 practice, finite-size effects, fabrication imperfections, and material losses limit Q to 10^2 – 10^4 .

579 Simulation of BIC-based sensors requires FDTD or eigenmode solvers (e.g., MPB, COMSOL)
 580 with fine frequency resolution and careful convergence testing, as the resonance width can be
 581 narrower than 0.1 nm. Sensitivity analysis—tracking the resonance shift $\Delta\lambda$ per refractive index
 582 unit (RIU) change—involves repeated simulations with perturbed background index, making
 583 computational efficiency a priority. TCMT (Section 3.4) provides a powerful complementary
 584 framework for rapid sensitivity estimation once the modal parameters are known.

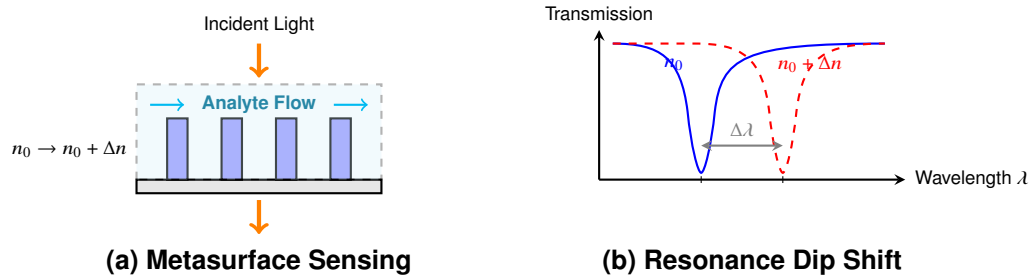


Fig. 16. Schematic of metasurface-based refractive index sensing. (a) An analyte of refractive index n_0 flowing over a high- Q resonant metasurface shifts the local index to $n_0 + \Delta n$. (b) This perturbation induces a shift $\Delta\lambda$ in the sharp transmission resonance, enabling high-sensitivity detection.

585 4.7. Computational Metaoptics

586 Computational metaoptics represents the frontier convergence of flat optics, machine learning,
 587 and computational imaging [36–39]. In this paradigm, the metasurface is not designed to produce
 588 a perfect image or beam, but rather to encode *information* that is decoded by a downstream
 589 computational algorithm. Examples include: depth estimation from a single image using a
 590 specially designed PSF, object classification performed optically through a diffractive neural
 591 network, and extended depth-of-field imaging using a coded aperture metalens.

592 The simulation framework for computational metaoptics is inherently *end-to-end*: a differ-
 593 entiable wave-optics model (ASM or angular spectrum in PyTorch/JAX) computes the image
 594 formed by the metalens on a detector, and a neural network processes this image to extract the
 595 desired information. Both the metasurface phase profile and the neural network weights are
 596 jointly optimized by backpropagation through the entire pipeline. This co-design approach has
 597 produced metalenses whose combined optical–computational performance significantly exceeds
 598 that of either component optimized independently, marking a paradigm shift from “perfect optics”
 599 to “optimal information” design [9].

600 5. Limitations and Challenges

601 5.1. Fundamental Simulation Limitations

602 Several fundamental limitations constrain the fidelity and scope of metaoptics simulation. The
 603 *scalability wall*—the inability of full-wave methods to handle device sizes beyond $\sim 100\ \mu\text{m}$ —
 604 forces reliance on the LPA, whose accuracy degrades systematically for high-NA devices,
 605 rapidly varying phase profiles, and meta-atoms with strong near-field coupling. Quantifying this
 606 degradation remains difficult: no standardized benchmark compares LPA-predicted performance

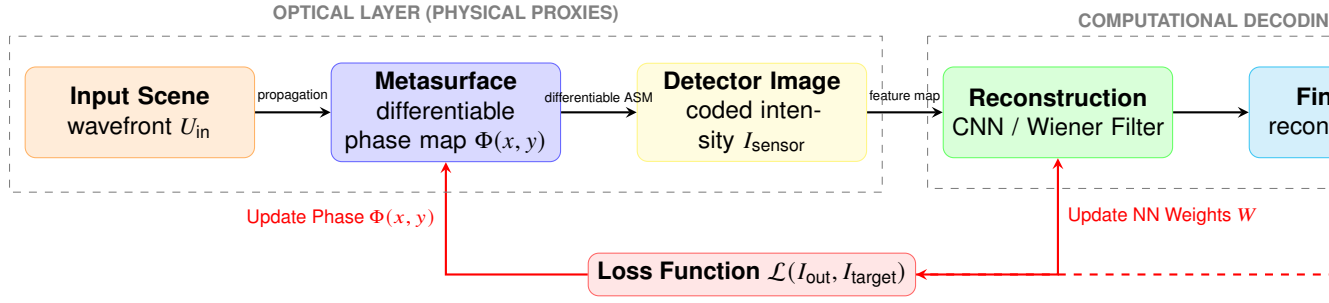


Fig. 17. End-to-end differentiable co-design pipeline. The physical optical layer (metasurface transmission and wave propagation) is modeled using differentiable physics-based operations (like ASM) to produce a coded detector image. The digital layer reconstructs the target image using a neural network. The design is optimized jointly by backpropagating the loss through both the digital network and the physical optical layer.

Table 5. Application domains and the typical deployment of simulation methods. (P = Primary method, S = Secondary/Validation method, E = Emerging method, – = Rarely used or N/A).

Application Domain	Full-Wave (FDTD/RCWA)	Fourier/ASM	Ray Tracing	ML Surrogate
Metalenses	S (Unit cell/Small lens)	P (Propagation)	S (System design)	E (Surrogate model)
Beam Shaping	S (Unit cell/Small aperture)	P (Propagation)	–	E (Fast search)
Holography	S (Unit cell library)	P (GS algorithm/ASM)	–	E (Diffraction simulation)
Achromatic	P (Spectral sweeps)	P (Wavelength sweep)	S (Multi-wavelength)	P (Dispersion model)
Tunable	P (Multi-physics EM)	S (State propagation)	–	E (State search)
Sensing/BIC	P (Resonance tracking)	–	–	E (Resonance model)
Computational	S (Unit cell library)	P (Differentiable ASM)	–	P (Differentiable model)

607 against full-wave ground truth across a range of device parameters. *Material model accuracy* is
608 another concern, particularly for phase-change materials, doped semiconductors, and anisotropic
609 media, where broadband dispersive fits may deviate from measured optical constants. The
610 *periodic–aperiodic mismatch*—the use of periodic boundary conditions to simulate meta-atoms
611 that are, in the assembled device, surrounded by non-identical neighbors—introduces a systematic
612 error that is difficult to quantify without prohibitively expensive supercell simulations.

613 5.2. The Fabrication–Simulation Gap

614 Perhaps the most significant—and least discussed—source of discrepancy between simulated
615 and measured metasurface performance is the *fabrication–simulation gap*. Simulations assume
616 idealized geometries: perfectly vertical sidewalls, atomically smooth surfaces, exact dimensions,
617 and uniform material composition. Real fabrication introduces deviations including sidewall
618 tapering (typically 2–10° from vertical), corner rounding (radii of 5–20 nm), line-edge roughness
619 (1–5 nm RMS), residual resist, surface oxidation, and film thickness non-uniformity. These
620 imperfections can reduce focusing efficiency by 10–30%, shift resonance wavelengths, and

621 degrade polarization purity.

622 Incorporating fabrication constraints *a priori* into the simulation and optimization workflow
 623 is an active area of research. Approaches include adding random geometric perturbations to
 624 simulated structures (Monte Carlo robustness analysis), encoding minimum feature sizes and
 625 spacing as constraints in topology optimization, and using SEM images of fabricated structures
 626 as inputs to FDTD for realistic performance prediction. However, no standardized methodology
 627 for fab-aware simulation exists, and most published metalens demonstrations do not report
 628 fabrication-induced performance degradation.

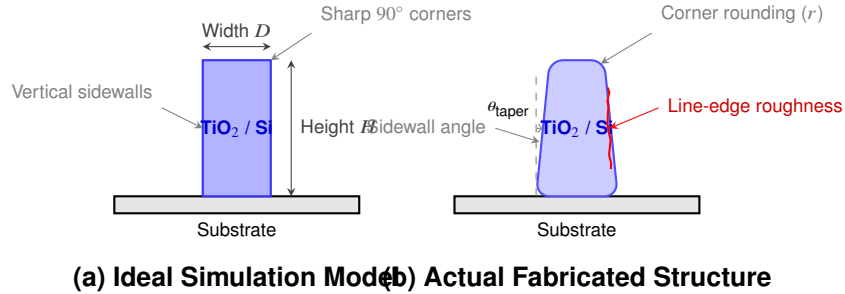


Fig. 18. The fabrication-simulation gap in metaoptics. (a) In simulations, meta-atoms are modeled with ideal geometries (e.g., vertical sidewalls, sharp corners, exact dimensions). (b) Fabricated nanostructures display systematic variations, including sidewall tapering (θ_{taper}), corner rounding (r), and line-edge roughness, which degrade device efficiency and optical quality.

629 5.3. Experimental Validation Challenges

630 Experimental validation of metasurface simulations faces its own challenges. Most simulation
 631 metrics—complex electric field amplitude and phase, near-field distributions, vectorial far-field
 632 patterns—are difficult to measure directly. Standard characterization relies on intensity-based
 633 measurements (focal spot imaging, far-field diffraction patterns), which provide incomplete
 634 information about the underlying field. *Efficiency* measurements are particularly problematic:
 635 the distinction between absolute efficiency (power in focal spot / total incident power), relative
 636 efficiency (power in focal spot / power transmitted through substrate), and diffraction efficiency
 637 (power in desired order / total diffracted power) is often ambiguous or inconsistently applied
 638 across publications, making fair comparison between simulated and measured results—and
 639 between different research groups—challenging.

640 5.4. Computational Cost and Accessibility

641 The computational cost of metaoptics simulation creates significant accessibility barriers.
 642 Commercial FDTD licenses (Lumerical, CST) cost \$15,000–\$50,000 per year, placing them
 643 beyond the reach of many academic groups, particularly in developing countries. Open-source
 644 alternatives (MEEP, S4) are functional but lack the graphical interfaces, optimization tools,
 645 and technical support of commercial packages. GPU acceleration is emerging as a potential
 646 equalizer: cloud-based services (Tidy3D) offer pay-per-simulation pricing that may be more
 647 accessible than annual licenses, and differentiable simulation frameworks in PyTorch/JAX can
 648 leverage commodity GPUs. However, the absence of community-wide *standardized benchmarks*—
 649 equivalent to ImageNet for computer vision or GLUE for NLP—means that simulation accuracy,
 650 speed, and cost cannot be objectively compared across tools and methods, hindering informed
 651 method selection and reproducibility.

Table 6. Summary of key computational and physical limitations for each simulation methodology.

Method	Computational Bottleneck	Physical Approximations	Priority
FDTD	$O(N^4)$ time, $O(N^3)$ memory scaling	Grid discretization (staircasing)	Medium
RCWA	Scaling with number of Fourier harmonics	Periodic boundary conditions assumed	Low
Fourier/ASM	FFT sampling requirements at large grids	Scalar diffraction, local phase approx. (LPA)	Intermediate
Ray Tracing	Ray tracing iteration overhead	Geometrical optics limit (no diffraction)	Fair
ML Surrogate	Upfront training dataset generation cost	Black-box data fitting, limited extrapolation	Considerable
Adjoint/TO	Multiple full-wave simulations per iteration	Staircasing, fabrication grid mismatched	Low
Hybrid	Interface domain-matching computational overhead	LPA or domain decoupling approximations	Erroneous

6. Future Outlook and Emerging Directions

Several emerging directions promise to reshape the computational landscape of metaoptics in the coming years. *Quantum metaoptics*—metasurfaces designed to manipulate quantum states of light (single photons, entangled pairs, squeezed states)—introduces new simulation requirements, including the need to model photon statistics and quantum coherence alongside classical electromagnetic fields. *Nonlinear metasurfaces* exploiting $\chi^{(2)}$ and $\chi^{(3)}$ processes (second-harmonic generation, four-wave mixing, optical parametric amplification) demand time-domain simulations with field-dependent material models, significantly increasing computational complexity. *3D and volumetric metaoptics*—multi-layer metasurface stacks and metaoptic elements with axial structure—expand the design space but require full 3D simulation or generalized transfer-matrix approaches.

On the computational side, *foundation models for photonic design*—large neural networks pre-trained on diverse electromagnetic simulation datasets and fine-tuned for specific tasks—could dramatically reduce the per-problem training cost of ML-based design. *Fully differentiable end-to-end pipelines* (Section 3.4), once mature, will unify design, simulation, and application optimization in a single framework. Most critically, the development of *standardized simulation benchmarks*—curated datasets of metaoptical designs with both full-wave ground truth and experimental validation data—would enable objective comparison of simulation methods and accelerate progress across the field.

7. Conclusion

This review has provided a systematic and comparative analysis of computational methods for metaoptics, covering the full spectrum from rigorous full-wave solvers to data-driven machine learning approaches. Our central finding is that *no single simulation method is adequate for all metaoptics design tasks*: the choice of method is dictated by the device scale, required accuracy, available computational resources, and target application. Based on our analysis, we recommend the following practical guidelines:

- **Small devices (<100 μm):** Full-wave FDTD simulation captures all relevant physics, including near-field coupling and edge effects. This is the most reliable approach but is limited by computational cost.
- **Medium devices (100 μm – 1 mm):** The standard FDTD + ASM hybrid workflow provides

- a practical balance of accuracy and speed, with the LPA as the primary source of error.
- **Large devices (>1 mm):** ML surrogates or RCWA for unit-cell libraries, combined with ASM propagation, are the only tractable approaches. Accuracy depends critically on surrogate fidelity and LPA validity.
 - **System-level design:** Ray tracing with metasurfaces modeled as thin phase elements enables multi-element system optimization, complemented by wave-optics verification of the final design.
 - **Resonant and sensing devices:** Full-wave eigenmode solvers or FDTD with fine spectral resolution, complemented by TCMT for physical insight.
 - **Computational co-design:** End-to-end differentiable pipelines that jointly optimize optics and algorithms.

We close with a call to action for the metaoptics community: the development of *open-source, standardized simulation benchmarks*—including reference designs with full-wave ground truth, fabricated device measurements, and documented computational costs—would transform the field by enabling objective method comparison, accelerating the adoption of new techniques, and improving reproducibility. The remarkable progress of the past decade, from the first metasurface demonstrations to commercial metalens products, has been enabled by computational methods. The next decade’s advances will be similarly shaped by the simulation tools we build today.

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